

QF620 Stochastic Modelling in Finance — Session 1

Reference Cheatsheet

Probability Theory and Lattice Models: expectation, moments, MGFs, normal and lognormal, two-state and three-state option pricing

Built from Session 1 slides (Tee Chyng Wen, SMU) · September 2026

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Source: “Session 1: Probability Theory and Lattice Models” (Tee Chyng Wen, SMU, 22 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. $\triangle!$ marks a slide ambiguity or error, with the corrected version.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From this session you need: the definitions of mean, variance, moments and central moments; skewness and kurtosis; the MGF and how to get moments from it; the normal MGF (and its derivation by completing the square); lognormal expectations via the MGF; and the two-state hedging argument and three-state bounds by hand.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
X	random variable	random	continuous unless stated
$f(x)$	probability density function (PDF)	function	“contains all information” about X [S2]
$F(x)$	cumulative distribution function (CDF)	function	$F(x) = P(X \leq x)$; used for median/quartiles [S4]
$\mathbb{E}[g(X)]$	expectation of $g(X)$	number	$\int g(x)f(x) dx$ [S2]
μ	mean $\mathbb{E}[X]$	number	first moment
$\mathbb{V}[X], \sigma^2$	variance $\mathbb{E}[(X - \mu)^2]$	number	slides write $V[X]$
σ	standard deviation	number	$\sqrt{\mathbb{V}[X]}$
$\mathbb{E}[X^k]$	k -th (raw) moment	number	about the origin [S6]
ν_k	k -th central moment $\mathbb{E}[(X - \mu)^k]$	number	about the mean [S7]; $\nu_1 = 0, \nu_2 = \sigma^2$
γ_k	standardised moment ν_k/σ^k	dimensionless	γ_3 skewness, γ_4 kurtosis [S9]
$M_X(\theta)$	moment generating function $\mathbb{E}[e^{\theta X}]$	function of θ	$M_X(0) = 1$ [S10]
$\phi(x)$	standard normal PDF	function	$\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ [S11]
$N(\mu, \sigma^2)$	normal distribution	—	second argument is the variance
$Y = e^X$	lognormal variable	random	$\ln Y \sim N(\mu, \sigma^2)$ [S14]
S_0, S_1	stock price today, tomorrow	price	100 today; 110 or 90 tomorrow [S15]
C_0, C_1	call value today, tomorrow	price	strike $K = 100$
Δ	shares held in the hedge	number	portfolio = $\Delta \times \text{stock} - 1 \times \text{call}$ [S16]
p^*	risk-neutral probability of the up state	probability	$p^* = \frac{1}{2}$ in the two-state example [S18]
p	real-world (empirical) probability	probability	irrelevant for the no-arbitrage price [S17, S19]

Notation trap: ν_k (Greek nu, central moment) versus v ; [S9] writes “ v_4 ” for ν_4 . Also γ_4 is kurtosis (normal = 3), while *excess* kurtosis is $\gamma_4 - 3$.

0. Topic map

PREREQUISITES

Undergraduate probability & calculus: PDFs, integration by substitution, Taylor series of e^x



SESSION 1: PROBABILITY THEORY AND LATTICE MODELS

└ DISTRIBUTIONS [S2–S5]

- └ $\mathbb{E}[g(X)] = \int g(x) f(x) dx$; $\mathbb{E}[a] = a$, $\mathbb{E}[aX] = a\mathbb{E}[X]$, linearity; $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ [S2]
- └ $\mu = \mathbb{E}[X]$; $\mathbb{V}[X] = \mathbb{E}[(X - \mu)^2]$; $\mathbb{V}[a] = 0$; $\mathbb{V}[aX + b] = a^2\mathbb{V}[X]$; $\sigma = \sqrt{\mathbb{V}}$ [S3]
- └ Mode (peak of f), median $F(M) = \frac{1}{2}$, quartiles $\frac{1}{4}, \frac{3}{4}$, percentiles; example [S4–S5]

└ MOMENTS & MGF [S6–S10]

- └ Raw moments $\mathbb{E}[X^k]$; $\mathbb{V}[X] = \mathbb{E}[X^2] - \mu^2$ [S6]
- └ Central moments $\nu_k = \mathbb{E}[(X - \mu)^k]$; $\nu_1 = 0$, $\nu_2 = \sigma^2$ [S7]
- └ $\nu_3 = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3$ [S8]
- └ $\gamma_k = \nu_k/\sigma^k$: skewness γ_3 , kurtosis γ_4 (normal = 3), excess = $\gamma_4 - 3$ [S9]
- └ $M_X(\theta) = \mathbb{E}[e^{\theta X}]$; $\mathbb{E}[X^n] = M_X^{(n)}(0)$ [S10]

└ NORMAL & LOGNORMAL [S11–S14]

- └ $\int e^{-y^2} dy = \sqrt{\pi} \Rightarrow$ normal PDF integrates to 1 [S11]
- └ Completing the square: $M_X(\theta) = \exp(\mu\theta + \frac{1}{2}\sigma^2\theta^2)$; $M_Z(\theta) = e^{\theta^2/2}$ [S12–S13]
- └ $Y = e^X$ lognormal: $\mathbb{E}[Y] = M_X(1) = e^{\mu + \sigma^2/2}$; returns normal, prices lognormal [S14]

└ LATTICE MODELS [S15–S22]

└ Two-state: $100 \rightarrow 110/90$, call $K=100$, $r=0$	[S15]
└ Hedge $\Delta=\frac{1}{2}$ makes portfolio worth 45 in both states $\Rightarrow C_0 = 5$	[S16]
└ No probability used; $p=0$ or 1 would be an arbitrage	[S17]
└ Risk-neutral $p^* = \frac{1}{2}$: the only p reproducing $C_0 = 5$; $E^*[S_1] = S_0$	[S18]
└ Real-world p irrelevant: hedging removes the risk premium	[S19]
└ Three-state: $110/100/90$; one Δ , two constraints $\Rightarrow$ cannot hedge	[S20–S21]
└ Risk-neutral $p^* \in (0, \frac{1}{2}) \Rightarrow 0 < C_0 < 5$: incomplete market, price interval	[S22]

↓

LATER: S2 multi-period binomial trees, $r \neq 0$, Radon-Nikodym, martingale pricing;
S3 Brownian motion as the limit of a random walk (CLT); S4 Itô calculus

Dependency logic:

- Expectation (linearity) gives variance and all moment identities.
- The MGF packages every moment into one function; for the normal it has a closed form, which then gives lognormal expectations for free ($\mathbb{E}[e^X] = M_X(1)$). This is the tool used again in S2 ($\mathbb{E}[S_T] = S_0 e^{rT}$) and S3 ($\mathbb{E}[e^{\sigma W_t}]$).
- The lattice half is the economic core of the course: replication/no-arbitrage fixes prices, and “risk-neutral probabilities” are just the weights that reproduce those prices.

1 Concept entries

1.1 Expectation and its properties [S2]

- **Definition** [S2]: $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$. The PDF “contains all information” about X .
- **Properties** [S2]:
 1. $\mathbb{E}[a] = a$ for a constant a .
 2. $\mathbb{E}[ag(X)] = a\mathbb{E}[g(X)]$.
 3. Linearity: $\mathbb{E}[s(X) + t(X)] = \mathbb{E}[s(X)] + \mathbb{E}[t(X)]$.
- **In general** $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ [S2]. *[Beyond slides]* Equality holds when X, Y are independent (more generally, uncorrelated): $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \text{Cov}(X, Y)$.
- **Discrete version** *[Beyond slides]*: $\mathbb{E}[g(X)] = \sum_x g(x)P(X = x)$ (used in the lattice models).
- **Traps:**
 - $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$ for non-linear g (e.g. $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$; $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$ by Jensen *[Beyond slides]*).
 - Linearity needs no independence; the product rule does.

1.2 Mean, variance and standard deviation [S3]

- **Definitions** [S3]: $\mu = \mathbb{E}[X] = \int x f(x) dx$; $\mathbb{V}[X] = \mathbb{E}[(X - \mu)^2] = \int (x - \mu)^2 f(x) dx$; $\sigma = \sqrt{\mathbb{V}[X]}$.
- **Properties** [S3]: $\mathbb{V}[a] = 0$; $\mathbb{V}[aX + b] = a^2 \mathbb{V}[X]$ (shifts do nothing; scaling enters squared).
- **Shortcut** [S6]: $\mathbb{V}[X] = \mathbb{E}[X^2] - \mu^2$. *[Beyond slides]* Proof: expand $(X - \mu)^2$ and use linearity: $\mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2$.
- **Worked example** *[Beyond slides]*: X uniform on $[0, 1]$: $\mathbb{E}[X] = \frac{1}{2}$, $\mathbb{E}[X^2] = \frac{1}{3}$, $\mathbb{V}[X] = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. Then $\mathbb{V}[3X + 2] = 9/12 = 0.75$.
- **⚠** [S3] says “the variance is always positive”. It is always *non-negative*: $\mathbb{V}[a] = 0$ for a constant, as property 1 on the same slide shows.
- **Traps:** $\mathbb{V}[aX + b] = a\mathbb{V}[X]$ (forgot the square); $\mathbb{V}[X - Y] = \mathbb{V}[X] - \mathbb{V}[Y]$ (it is $\mathbb{V}[X] + \mathbb{V}[Y] - 2\text{Cov}$ *[Beyond slides]*).

1.3 Mode, median, quartiles and percentiles [S4–S5]

- **Mode** [S4]: the value of x where $f(x)$ is greatest (there may be several).
- **Median** [S4]: M with $F(M) = \frac{1}{2}$. Quartiles: $F(Q_{\text{lower}}) = \frac{1}{4}$, $F(Q_{\text{upper}}) = \frac{3}{4}$; they split the distribution into four regions of equal probability. Percentile P_n : $F(P_n) = n/100$.
- Neither requires computing an expectation [S4].
- **Slide example** [S5]: temperature over 110 days: 80 for 100 days, 85 for 5 days, 90 for 5 days. Median?
 - *[Beyond slides]* Sorted, positions 55 and 56 are both 80, so the median is **80**. The mode is also 80.
 - The mean is $(100 \cdot 80 + 5 \cdot 85 + 5 \cdot 90)/110 = 80.68$: pulled up by the high days. The median ignores how extreme the tail values are.
- **⚠** $F(M) = \frac{1}{2}$ may have no solution for a discrete distribution (the CDF jumps over $\frac{1}{2}$, as in [S5]). General definition *[Beyond slides]*: the smallest x with $F(x) \geq \frac{1}{2}$.

- **Connection:** lognormal prices have mode < median < mean (1.9).

1.4 Raw moments and central moments [S6–S8]

- **Raw moments** [S6]: $\mathbb{E}[X^k] = \int x^k f(x) dx$; the mean is the first.
- **Central moments** [S7]: $\nu_k = \mathbb{E}[(X - \mu)^k]$, moments “about the mean”. $\nu_1 = 0$ always; $\nu_2 = \sigma^2$.
- **Converting** [S8] (binomial expansion + linearity):

$$\nu_3 = \mathbb{E}[(X - \mu)^3] = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3.$$

[Beyond slides] Derivation: $\mathbb{E}[X^3] - 3\mu\mathbb{E}[X^2] + 3\mu^2\mathbb{E}[X] - \mu^3 = \mathbb{E}[X^3] - 3\mu\mathbb{E}[X^2] + 2\mu^3$.

- [Beyond slides] Similarly $\nu_4 = \mathbb{E}[X^4] - 4\mu\mathbb{E}[X^3] + 6\mu^2\mathbb{E}[X^2] - 3\mu^4$.
- **Worked example** [Beyond slides]: exponential(1) has $\mathbb{E}[X^k] = k!$, so $\nu_3 = 6 - 3(1)(2) + 2(1) = 2$ and $\gamma_3 = 2/1^3 = 2$ (right-skewed).
- **Traps:** writing $3\mathbb{E}[X^2]\mathbb{E}[X]$ with the wrong sign; forgetting the $+2\mu^3$.

1.5 Skewness and kurtosis [S9]

- **Standardised moment** [S9]: $\gamma_k = \nu_k/\sigma^k$ (dimensionless).
- **Skewness** γ_3 : 0 for a symmetric distribution; $\gamma_3 < 0$ if skewed toward values below the mean (long left tail), > 0 for a long right tail.
- **Kurtosis** γ_4 : for $N(\mu, \sigma^2)$, $\gamma_4 = \nu_4/\sigma^4 = 3\sigma^4/\sigma^4 = 3$.
- **Excess kurtosis** $= \gamma_4 - 3$: positive means a narrower peak and fatter tails than the normal with the same mean and variance (“leptokurtic”).
- **Finance reading** [Beyond slides]: equity index returns typically show negative skew and positive excess kurtosis (crashes); the normal model underestimates tail risk.
- **△!** [S9] writes “ ν_4 ” in the kurtosis formula; it is ν_4 .
- **Traps:** quoting 3 as “excess kurtosis of the normal” (it is 0); mixing up the sign convention of skew.

1.6 Moment generating functions [S10]

- **Definition** [S10]: $M_X(\theta) = \mathbb{E}[e^{\theta X}] = \int e^{\theta x} f(x) dx$; $M_X(0) = 1$.
- **Why it generates moments** [S10]: expand $e^{\theta X}$ and take expectations term by term:

$$M_X(\theta) = 1 + \mathbb{E}[X]\theta + \mathbb{E}[X^2]\frac{\theta^2}{2!} + \mathbb{E}[X^3]\frac{\theta^3}{3!} + \cdots \Rightarrow \mathbb{E}[X^n] = \left. \frac{d^n M_X}{d\theta^n} \right|_{\theta=0}.$$

- **Two routes** [Beyond slides]: differentiate n times at 0, or expand $M_X(\theta)$ as a power series and read off $n! \times (\text{coefficient of } \theta^n)$. The second is usually faster.
- **Useful facts** [Beyond slides]:
 - $M_{aX+b}(\theta) = e^{b\theta} M_X(a\theta)$.
 - Independent X, Y : $M_{X+Y} = M_X M_Y$.
 - The MGF (when it exists near 0) determines the distribution.
- **Worked example** [Beyond slides] (normal moments from $M_X = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$):
 - $M' = (\mu + \sigma^2\theta)M \Rightarrow \mathbb{E}[X] = \mu$.
 - $M'' = [\sigma^2 + (\mu + \sigma^2\theta)^2]M \Rightarrow \mathbb{E}[X^2] = \sigma^2 + \mu^2$, so $\mathbb{V} = \sigma^2 \checkmark$.
 - Standard normal: $e^{\theta^2/2} = \sum_k \frac{\theta^{2k}}{2^k k!}$, so $\mathbb{E}[Z^{2k}] = \frac{(2k)!}{2^k k!}$: $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, $\mathbb{E}[Z^6] = 15$; odd moments 0. Hence $\nu_4 = 3\sigma^4$ and $\gamma_4 = 3 \checkmark$.
- **Traps:** evaluating the derivative at $\theta = 1$ instead of 0; confusing the MGF with the characteristic function $\mathbb{E}[e^{i\theta X}]$ [Beyond slides].

1.7 The normal PDF [S11]

- **Gaussian integral** [S11]: $I = \int_{-\infty}^{\infty} e^{-y^2} dy$ satisfies $I^2 = \pi$, so $I = \sqrt{\pi}$. ([Beyond slides] I^2 is a double integral; switch to polar coordinates: $\int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\vartheta = \pi$.)
- **Standard normal** [S11]: substitute $y = x/\sqrt{2}$: $\int \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 1$, so $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is a density.
- **General normal** [S11]: substitute $y = (x - \mu)/(\sqrt{2}\sigma)$: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **Standardising** [Beyond slides]: $X \sim N(\mu, \sigma^2) \iff Z = (X - \mu)/\sigma \sim N(0, 1)$; $P(X \leq x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$.
- **Trap:** the normalising constant is $\sqrt{2\pi}\sigma$, not $\sqrt{2\pi}\sigma$ or $\sqrt{2\pi}\sigma^2$.

1.8 MGF of the normal distribution [S12–S13]

- **Result** [S12–S13]: $X \sim N(\mu, \sigma^2) \Rightarrow M_X(\theta) = \exp(\mu\theta + \frac{1}{2}\sigma^2\theta^2)$; for $Z \sim N(0, 1)$, $M_Z(\theta) = e^{\theta^2/2}$.
- **Derivation by completing the square** [S12–S13]:
 1. Combine exponents: $\theta x - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{x^2 - 2(\mu + \sigma^2\theta)x + \mu^2}{2\sigma^2}$.
 2. Add and subtract $(\mu + \sigma^2\theta)^2$: numerator = $[x - (\mu + \sigma^2\theta)]^2 - 2\mu\sigma^2\theta - \sigma^4\theta^2$.
 3. So the exponent is $-\frac{[x - (\mu + \sigma^2\theta)]^2}{2\sigma^2} + \mu\theta + \frac{\sigma^2\theta^2}{2}$.
 4. The remaining integral is the PDF of $N(\mu + \sigma^2\theta, \sigma^2)$, which integrates to 1.
- **Faster route** [Beyond slides]: $X = \mu + \sigma Z$, so $M_X(\theta) = e^{\mu\theta} M_Z(\sigma\theta)$; and M_Z needs only one completed square.
- **By-product** [Beyond slides] (used in S2–S4): multiplying a normal density by $e^{\theta x}$ shifts its mean by $\sigma^2\theta$. This is the seed of the change-of-measure idea.
- **Trap**: writing $e^{\mu\theta + \sigma^2\theta^2}$ (missing $\frac{1}{2}$) or $e^{\mu\theta + \frac{1}{2}\sigma\theta^2}$ (variance vs s.d.).

1.9 Lognormal random variables [S14]

- **Definition** [S14]: $X \sim N(\mu, \sigma^2)$, $Y = e^X$ is lognormal, because $\ln Y \sim N(\mu, \sigma^2)$.
- **Moments from the normal MGF** [S14]: $\mathbb{E}[Y^n] = \mathbb{E}[e^{nX}] = M_X(n)$.
 - $\mathbb{E}[Y] = M_X(1) = e^{\mu + \frac{1}{2}\sigma^2}$ [S14].
 - [Beyond slides] $\mathbb{E}[Y^2] = e^{2\mu + 2\sigma^2}$, so $\mathbb{V}[Y] = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$.
 - [Beyond slides] median e^μ , mode $e^{\mu - \sigma^2}$: mode < median < mean.
- **Worked example** [Beyond slides]: $\mu = 0.05$, $\sigma = 0.2$: $\mathbb{E}[Y] = e^{0.07} = 1.0725$; median $e^{0.05} = 1.0513$. The $\frac{1}{2}\sigma^2$ gap is the “convexity” term that becomes $-\frac{1}{2}\sigma^2$ in $S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}$ (S2).
- **Slide question** [S14]: why model returns as normal and prices as lognormal? [Beyond slides] answer:
 - Prices must stay positive; $e^X > 0$ always, while a normal price can go negative.
 - Log-returns add over time: $\ln(S_T/S_0) = \sum \ln(S_{t_i}/S_{t_{i-1}})$. A sum of many independent small returns is approximately normal (CLT, S2), so the price is lognormal.
 - Limited liability: a stock can lose at most 100% but gain without bound; the lognormal’s right skew matches this.
- **Traps**: $\mathbb{E}[e^X] = e^{\mathbb{E}X}$ (wrong by the factor $e^{\sigma^2/2}$); treating μ, σ as the mean and s.d. of Y rather than of $\ln Y$.
- **Implementation** [Beyond slides] (tested):

```
import numpy as np
mu, sig = 0.05, 0.2
Y = np.exp(np.random.default_rng(0).normal(mu, sig, 1_000_000))
print(Y.mean(), np.exp(mu + sig**2 / 2)) # 1.0725 vs 1.0725
print(np.median(Y), np.exp(mu)) # 1.0513 vs 1.0513
```

1.10 Two-state model: pricing by hedging [S15–S16]

- **Setup** [S15]: stock 100 today, 110 or 90 tomorrow; ATM call $K = 100$; $r = 0$. Call payoff: 10 (up) or 0 (down), so $0 \leq C_0 \leq 10$.
- **Hedge** [S16]: sell the call, buy Δ shares: portfolio $\Delta \times \text{stock} - 1 \times \text{call}$. Make it state-independent:

$$\underbrace{110\Delta - 10}_{\text{up}} = \underbrace{90\Delta - 0}_{\text{down}} \Rightarrow \Delta = \frac{10 - 0}{110 - 90} = \frac{1}{2}.$$

- **Price** [S16]: portfolio is worth 45 in both states; riskless with $r = 0 \Rightarrow$ worth 45 today:

$$100 \cdot \frac{1}{2} - C_0 = 45 \Rightarrow C_0 = 5.$$

- **General formula** [Beyond slides]: $\Delta = \frac{C_u - C_d}{S_u - S_d}$ (the “delta” is the slope of the option value in the stock).
- **Arbitrage if mispriced** [Beyond slides]: if $C_0 = 6$, sell the call, buy $\frac{1}{2}$ share, borrow $50 - 6 = 44$; tomorrow the position is worth $45 - 44 = +1$ in every state.
- **Traps**: buying the call in the hedge portfolio (sign); forgetting to discount when $r \neq 0$ (S2).

1.11 Probabilities and the risk-neutral measure [S17–S19]

- **No probabilities needed** [S17]: the hedge argument holds whatever the up/down probabilities are.

- **But both states must be possible** [S17]:
 - $P(\text{up}) = 1$ is impossible: the stock would be guaranteed to beat the risk-free rate; borrow and buy stock for a riskless profit.
 - $P(\text{down}) = 1$ is impossible (slide “why?”; [Beyond slides]): the stock would be guaranteed to underperform; short the stock and invest the proceeds risk-free.
 - The probabilities’ only role is to say that both states can happen.
- **Risk-neutral probability** [S18]: choose p^* so that expectation reproduces the no-arbitrage price.
 - $\mathbb{E}^*[C_1] = 10p^* = C_0 = 5 \Rightarrow p^* = \frac{1}{2}$ (the only such value).
 - Check: $\mathbb{E}^*[S_1] = 110 \cdot \frac{1}{2} + 90 \cdot \frac{1}{2} = 100 = S_0$: with $r = 0$ the stock is expected to earn nothing.
 - Equivalent definition [Beyond slides]: p^* solves $\mathbb{E}^*[S_1] = S_0(1+r)$, i.e. $p^* = \frac{S_0(1+r)-S_d}{S_u-S_d}$ (S2).
- **Real-world vs risk-neutral** [S19]:
 - If an arbitrage exists for one $p \in (0,1)$ it exists for all; if none exists for one p , none exists for any. Arbitrage depends only on which states are possible.
 - Under p^* we demand no risk premium for holding the stock.
 - In the real world, risk aversion means $p > \frac{1}{2}$ here, so real-world expected values of stock and option exceed the arbitrage-free price.
 - “The ability to hedge has removed the risk premium.”
- **△!** [S19] “the risk-neutral price ...only uses zero and non-zero probabilities” means: the price depends only on *which* states have positive probability, not *how much*. This is the equivalence-of-measures idea formalised in S2.
- **△!** [S19] “one would expect $p \geq \frac{1}{2}$ ”: more precisely $p > p^*$ for a risk-averse market with a positive risk premium ($p^* = \frac{1}{2}$ only because $r = 0$ and the moves are symmetric).
- **Traps:** “ p^* is the market’s belief” (it is not; it is a pricing device); using p to price and getting a price that admits arbitrage.

1.12 Three-state model and incomplete markets [S20–S22]

- **Setup** [S20]: add a middle state: 110, 100, 90 with call payoffs 10, 0, 0.
- **Hedging fails** [S21]: the portfolio is worth $110\Delta - 10$, 100Δ , 90Δ . One unknown, two constraints (“which two?” [Beyond slides]: up = middle and middle = down; the third equality then follows). No Δ equalises all three.
 - Match middle and down: $\Delta = 0$, no hedge at all.
 - Match up and down: $\Delta = \frac{1}{2}$, portfolio worth 45, 50, 45: no longer riskless.
- **Upper bound from super-hedging** [S21]: the portfolio is worth at least 45 in every state and more in one, so today $50 - C_0 > 45 \Rightarrow C_0 < 5$.
- **Risk-neutral approach** [S22]: need $\mathbb{E}^*[S_1] = 100$, so $P^*(110) = P^*(90) = p^*$ and $P^*(100) = 1 - 2p^*$, any $0 < p^* < \frac{1}{2}$.
 - $C_0 = \mathbb{E}^*[C_1] = 10p^* \in (0, 5)$.
 - The set of arbitrage-free prices is an interval, $0 < C_0 < 5$, not a single number.
- **Incomplete market** [S22]: payoffs cannot be replicated exactly; prices are only bounded. [Beyond slides] Two instruments (stock, cash) cannot span three states; adding a third traded asset (e.g. another option) would complete the market.
- **Lower bound** [Beyond slides]: $C_0 > 0$ because the payoff is non-negative and positive with positive probability.
- **Traps:** claiming $C_0 = 5$ in the trinomial model; allowing $p^* = 0$ or $\frac{1}{2}$ (then a state has zero probability, contradicting equivalence).
- **Implementation** [Beyond slides] (tested):

```
def two_state(S0, Su, Sd, Cu, Cd, r=0.0):
    delta = (Cu - Cd) / (Su - Sd)
    p = (S0 * (1 + r) - Sd) / (Su - Sd) # risk-neutral up prob
    C0 = (p * Cu + (1 - p) * Cd) / (1 + r)
    return delta, p, C0
print(two_state(100, 110, 90, 10, 0)) # (0.5, 0.5, 5.0)
```

2 Comparison tables

2.1 Measures of location and spread

Measure	Definition	Needs an expectation?	Lognormal e^X
Mean μ	$\mathbb{E}[X]$	yes	$e^{\mu+\sigma^2/2}$
Median M	$F(M) = \frac{1}{2}$	no	e^μ
Mode	$\arg \max f(x)$	no	$e^{\mu-\sigma^2}$
Variance	$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2$	yes	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
Quartiles	$F(Q) = \frac{1}{4}, \frac{3}{4}$	no	$e^{\mu \pm 0.674\sigma}$

2.2 Raw moments, central moments, standardised moments

Order	Raw $\mathbb{E}[X^k]$	Central ν_k	Standardised γ_k (normal)
1	μ	0	—
2	$\sigma^2 + \mu^2$	σ^2	1
3	$\mathbb{E}[X^3]$	$\mathbb{E}[X^3] - 3\mu\mathbb{E}[X^2] + 2\mu^3$	skewness (0)
4	$\mathbb{E}[X^4]$	$\mathbb{E}[X^4] - 4\mu\mathbb{E}[X^3] + 6\mu^2\mathbb{E}[X^2] - 3\mu^4$	kurtosis (3)

2.3 Normal vs lognormal

	Normal $X \sim N(\mu, \sigma^2)$	Lognormal $Y = e^X$
Support	$(-\infty, \infty)$	$(0, \infty)$
Symmetry	symmetric, skew 0	right-skewed
MGF / moments	$M_X(\theta) = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$	$\mathbb{E}[Y^n] = M_X(n) = e^{n\mu + \frac{1}{2}n^2\sigma^2}$
Arises from	sums of many small shocks (CLT)	products of many positive shocks
Used for	log-returns	prices [S14]

2.4 Two-state vs three-state model

	Two-state [S15–S19]	Three-state [S20–S22]
States	110, 90	110, 100, 90
Unknowns vs constraints	Δ : 1 vs 1	Δ : 1 vs 2
Perfect hedge	yes, $\Delta = \frac{1}{2}$	no
Risk-neutral p^*	unique, $\frac{1}{2}$	any $p^* \in (0, \frac{1}{2})$, middle $1 - 2p^*$
Call price	$C_0 = 5$ (unique)	$0 < C_0 < 5$ (interval)
Market	complete	incomplete

3 Master formula sheet

Expectation and variance [S2–S3, S6]

- $\mathbb{E}[g(X)] = \int g(x)f(x) dx$; $\mathbb{E}[a] = a$; $\mathbb{E}[aX] = a\mathbb{E}[X]$; linearity.
- $\mathbb{V}[X] = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2$; $\mathbb{V}[aX + b] = a^2\mathbb{V}[X]$.

Median and quantiles [S4]: $F(M) = \frac{1}{2}$; $F(Q) = \frac{1}{4}, \frac{3}{4}$; $F(P_n) = n/100$. Moments [S6–S9]

- $\nu_k = \mathbb{E}[(X - \mu)^k]$; $\nu_3 = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3$.
- $\gamma_k = \nu_k/\sigma^k$; normal: $\gamma_3 = 0$, $\gamma_4 = 3$; excess kurtosis $= \gamma_4 - 3$.

MGF [S10, S12–S13]

- $M_X(\theta) = \mathbb{E}[e^{\theta X}]$; $M_X(0) = 1$; $\mathbb{E}[X^n] = M_X^{(n)}(0)$.
- Normal: $e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$; standard normal: $e^{\theta^2/2}$.
- $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, $\mathbb{E}[Z^{2k}] = (2k)!/(2^k k!)$ [Beyond slides].

Normal PDF [S11]: $\int e^{-y^2} dy = \sqrt{\pi}$; $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$. **Lognormal [S14]:** $\mathbb{E}[e^X] = e^{\mu + \frac{1}{2}\sigma^2}$; $\mathbb{E}[e^{nX}] = e^{n\mu + \frac{1}{2}n^2\sigma^2}$ [Beyond slides]. **Lattice [S16–S22]**

- $\Delta = \frac{C_u - C_d}{S_u - S_d}$; riskless portfolio value $= \Delta S_u - C_u = \Delta S_d - C_d$.
- $p^* = \frac{S_0(1+r) - S_d}{S_u - S_d}$; $C_0 = \frac{p^*C_u + (1-p^*)C_d}{1+r}$ ($r = 0$ in Session 1).
- Trinomial (symmetric, $r = 0$): $P^*(\text{up}) = P^*(\text{down}) = p^* \in (0, \frac{1}{2})$; price interval.

4 Glossary

- **Arbitrage:** a trading strategy with zero cost, no chance of loss and a positive chance of profit.
- **At-the-money (ATM):** strike equals the current price ($K = S_0 = 100$).
- **Central moment ν_k :** $\mathbb{E}[(X - \mu)^k]$ [S7].
- **Complete market:** every payoff can be replicated with traded assets; unique prices.
- **CDF $F(x)$:** $P(X \leq x)$.
- **Excess kurtosis:** $\gamma_4 - 3$; positive = fat tails [S9].
- **Hedge ratio Δ :** shares held per option to remove risk [S16].
- **Incomplete market:** some payoffs cannot be replicated; prices lie in an interval [S22].
- **Kurtosis γ_4 :** ν_4/σ^4 ; normal = 3.
- **Lattice (tree) model:** discrete-time model with finitely many next-step states [S15].
- **Lognormal:** Y with $\ln Y$ normal [S14].
- **Median / mode:** 50th percentile / peak of the PDF [S4].
- **MGF:** $M_X(\theta) = \mathbb{E}[e^{\theta X}]$ [S10].
- **No-arbitrage price:** the only price at which no arbitrage exists [S16].
- **PDF $f(x)$:** density; $P(a < X < b) = \int_a^b f$.
- **Real-world (empirical) probability p :** the actual likelihood of states; irrelevant for pricing by replication [S19].
- **Replication:** building a portfolio whose payoff matches the derivative.
- **Risk-neutral probability p^* :** weights under which expected discounted payoffs equal no-arbitrage prices [S18].
- **Skewness γ_3 :** ν_3/σ^3 ; 0 if symmetric [S9].

5 Self-test

Questions

- Q1 (easy).** X uniform on $[0, 2]$. Find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathbb{V}[X]$ and $\mathbb{V}[5 - 3X]$.
- Q2 (easy, [S5]).** Temperatures over 110 days: 80 for 100 days, 85 for 5 days, 90 for 5 days. Find the median, mode and mean, and explain why they differ.
- Q3.** Show that $\nu_3 = \mathbb{E}[X^3] - 3\mathbb{E}[X]\mathbb{E}[X^2] + 2\mathbb{E}[X]^3$. Apply it to an exponential(1) variable ($\mathbb{E}[X^k] = k!$) and find its skewness.
- Q4.** Using $M_Z(\theta) = e^{\theta^2/2}$, find $\mathbb{E}[Z^3]$, $\mathbb{E}[Z^4]$ and $\mathbb{E}[Z^6]$. Deduce the kurtosis and excess kurtosis of $N(\mu, \sigma^2)$.
- Q5.** Derive $M_X(\theta)$ for $X \sim N(\mu, \sigma^2)$ by completing the square, then use it to find $\mathbb{E}[X]$ and $\mathbb{V}[X]$.
- Q6.** $\ln Y \sim N(0.05, 0.2^2)$. Find $\mathbb{E}[Y]$, $\mathbb{V}[Y]$, the median and the mode of Y .
- Q7.** Why are returns modelled as normal and prices as lognormal? Give two reasons.
- Q8 (two-state).** $S_0 = 100$, $S_1 \in \{120, 80\}$, $r = 0$. Price a call with $K = 110$ by (a) hedging, (b) risk-neutral probabilities. Show an arbitrage if the call trades at 6.
- Q9 (conceptual).** Why can the up-probability not be 1? Why doesn't the real-world probability enter the

price? What does “hedging removes the risk premium” mean?

Q10 (three-state). $S_0 = 100$, $S_1 \in \{110, 100, 90\}$, $r = 0$, call $K = 100$. Explain why no perfect hedge exists, derive the upper bound with $\Delta = \frac{1}{2}$, and find the full range of arbitrage-free prices by the risk-neutral method.

Solutions

S1. $\mathbb{E}[X] = 1$; $\mathbb{E}[X^2] = \int_0^2 x^2 \cdot \frac{1}{2} dx = \frac{4}{3}$; $\mathbb{V}[X] = \frac{4}{3} - 1 = \frac{1}{3}$; $\mathbb{V}[5 - 3X] = 9 \cdot \frac{1}{3} = 3$.

S2. Median 80 (positions 55–56 are 80); mode 80; mean $8875/110 = 80.68$. The mean is pulled up by the 10 hot days; the median and mode ignore how far the tail values sit.

S3. Expand: $\mathbb{E}[X^3 - 3\mu X^2 + 3\mu^2 X - \mu^3] = \mathbb{E}[X^3] - 3\mu\mathbb{E}[X^2] + 3\mu^3 - \mu^3$. Exponential(1): $\nu_3 = 6 - 3 \cdot 1 \cdot 2 + 2 = 2$; $\sigma = 1$, so $\gamma_3 = 2$.

S4. $e^{\theta^2/2} = 1 + \frac{\theta^2}{2} + \frac{\theta^4}{8} + \frac{\theta^6}{48} + \dots$. Odd moments 0; $\mathbb{E}[Z^4] = 4!/8 = 3$; $\mathbb{E}[Z^6] = 6!/48 = 15$. $X = \mu + \sigma Z$, so $\nu_4 = \sigma^4 \mathbb{E}[Z^4] = 3\sigma^4$, kurtosis 3, excess 0.

S5. See 1.8. $M'(0) = \mu$; $M''(0) = \sigma^2 + \mu^2$; $\mathbb{V} = \sigma^2$.

S6. $\mathbb{E}[Y] = e^{0.05+0.02} = 1.0725$; $\mathbb{V}[Y] = e^{0.1+0.04}(e^{0.04} - 1) = 1.1503 \times 0.04081 = 0.0469$ (s.d. 0.217); median $e^{0.05} = 1.0513$; mode $e^{0.05-0.04} = 1.0101$.

S7. (i) Prices must be positive; $e^X > 0$. (ii) Log-returns add over time, so their sum is approximately normal by the CLT, making the price lognormal. (Also: limited liability gives right skew.)

S8.

- Payoffs: 10 and 0. $\Delta = 10/40 = \frac{1}{4}$. Portfolio $\frac{1}{4}S_1 - C_1$: $30 - 10 = 20$ and $20 - 0 = 20$. So $25 - C_0 = 20 \Rightarrow C_0 = 5$.
- $p^* = (100 - 80)/40 = \frac{1}{2}$; $C_0 = \frac{1}{2} \cdot 10 = 5$ ✓.
- At 6: sell the call (+6), buy $\frac{1}{4}$ share (−25), borrow 19. Tomorrow: portfolio 20, repay 19: +1 in both states.

S9. If up were certain, borrow at r and buy the stock for a sure profit (arbitrage). The price comes from replication, which works in every state, so only which states are possible matters. Because the option's risk can be hedged away, nobody is paid for bearing it, so pricing uses probabilities under which the stock earns just r .

S10. Portfolio values $110\Delta - 10$, 100Δ , 90Δ : two independent equalities, one unknown. With $\Delta = \frac{1}{2}$: 45, 50, $45 \geq 45$, strictly greater in one state, so $50 - C_0 > 45 \Rightarrow C_0 < 5$. Risk-neutral: $P^*(110) = P^*(90) = p^*$, $0 < p^* < \frac{1}{2}$; $C_0 = 10p^* \in (0, 5)$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Expectation and its properties; $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$	1.1
S3	Mean, variance, standard deviation	1.2
S4	Mode, median, quartiles, percentiles	1.3
S5	Example: median temperature	1.3
S6	Raw moments; $\mathbb{V} = \mathbb{E}[X^2] - \mu^2$	1.4
S7	Central moments ν_k	1.4
S8	Example: ν_3 in terms of raw moments	1.4
S9	Standardised moments; skewness; kurtosis; excess kurtosis	1.5
S10	MGF; moments as derivatives at 0	1.6
S11	Gaussian integral; normal PDF	1.7
S12	Normal MGF: completing the square (part 1)	1.8
S13	Normal MGF: result; standard normal MGF	1.8
S14	Lognormal; $\mathbb{E}[Y] = M_X(1)$; why normal returns / lognormal prices	1.9
S15	Two-state model setup	1.10
S16	Hedge $\Delta = \frac{1}{2}$; $C_0 = 5$	1.10

Slide	Content	Section
S17	Probabilities not needed; up/down probability 1 impossible	1.11
S18	Risk-neutral $p^* = \frac{1}{2}$	1.11
S19	Real-world vs risk-neutral; hedging removes risk premium	1.11
S20	Three-state model setup	1.12
S21	Hedging fails; upper bound $C_0 < 5$	1.12
S22	Risk-neutral range $0 < C_0 < 5$; incomplete market	1.12

Appendix B. Flag summary ($\triangle!$)

- [S3] “Variance is always positive” $\rightarrow$ non-negative (0 for constants).
- [S4–S5] $F(M) = \frac{1}{2}$ may have no solution for a discrete distribution; use the smallest x with $F(x) \geq \frac{1}{2}$.
- [S9] “ v_4 ” should be ν_4 .
- [S17] “Why?” (down-probability 1) is not answered: short the stock and invest risk-free.
- [S19] “Only uses zero and non-zero probabilities” = the price depends only on which states are possible (equivalence, S2). “ $p \geq \frac{1}{2}$ ” is really $p > p^*$ under risk aversion.
- [S21] “Which two?” constraints: up = middle and middle = down.
- [S5, S14] Slide questions (median; why lognormal) are posed without answers; the answers here are my own.